Classical Field Theory — Exercise Sheet 27

Matrix compatibility, formal charges and Bäcklund kinks

Hints and answers

The first two problems assume a smooth real KdV solution $u : I \times \mathbb{R} \to \mathbb{R}$, with I an open time interval, $u_t = 6uu_x - u_{xxx}$ and spatial period $L_{\mathrm{per}} > 0$. This is a conditional hypothesis, not an existence assertion on the circle. The last problem instead uses a real field on the whole line.

Exercise 27.1 (Converting the scalar KdV pair to matrices). For fixed $k \in \mathbb{C} \setminus \{0\}$ consider

$$-\psi_{xx} + u\psi = k^2\psi, \qquad \psi_t = -4\psi_{xxx} + 6u\psi_x + 3u_x\psi.$$

Set $\Psi = (\psi, \psi_x)^{\mathsf{T}}$.

(a) Eliminate higher spatial derivatives to derive $\Psi_x = U\Psi$, $\Psi_t = V\Psi$. Compute every entry of $U_t - V_x + [U, V]$ for an arbitrary smooth u.

(b) For the assumed periodic KdV field, state exactly what Lecture 27's monodromy theorem proves for these matrices. Does it establish the Poisson commutativity or independence of the resulting traces?

Hint. Differentiate $\psi_{xx} = (u - k^2)\psi$ once before substituting into the temporal equation. Differentiate the resulting temporal first row once to obtain the second row.

Answer. The pair and its curvature are

$$U = \begin{pmatrix} 0 & 1 \\ u - k^2 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} -u_x & 2u + 4k^2 \\ -u_{xx} + (2u + 4k^2)(u - k^2) & u_x \end{pmatrix},$$

$$U_t - V_x + [U, V] = \begin{pmatrix} 0 & 0 \\ u_t - 6uu_x + u_{xxx} & 0 \end{pmatrix}.$$

Every trace power of its period transport is conserved. Independence and involution are not established by that result.

Exercise 27.2 (Local charges from a formal Riccati series). On a patch with $\psi \neq 0$, the scalar equation suggests $\rho = \psi_x/\psi$ and $\rho_x + \rho^2 = u - k^2$. In this problem work purely with formal Laurent series: put $\zeta = 2ik$ and seek

$$\rho = \frac{\zeta}{2} + \sum_{j \geq 1} a_j[u]\zeta^{-j}.$$

Equality means equality of each coefficient; convergence and a globally nonvanishing actual ψ are not assumed.

(a) Find the recursion and the first five coefficients. Integrate them over one spatial period and identify the first three nonzero integrals.

(b) Prove the formal identity $\rho_t = \partial_x[-u_x + (2u + 4k^2)\rho]$ under KdV. Conclude coefficientwise conservation. Explain which analytic conclusion about an actual spectral expansion has not been proved.

Hint. Subtract the leading $\zeta/2$ in the Riccati equation. For the second part set $S = -u_x + (2u + 4k^2)\rho$ and apply $\partial_x + 2\rho$ to $\rho_t - S_x$. A nonzero formal Laurent series bounded above in powers of ζ cannot be annihilated by $\partial_x + \zeta + O(\zeta^{-1})$.

Answer. The recursion is $a_1 = u$ and $a_{n+1} = -a_n' - \sum_{i+j=n,\ i,j\geq 1} a_i a_j$. The nonzero integrated coefficients through order five are

$$\int a_1 = \int u, \qquad \int a_3 = -\int u^2, \qquad \int a_5 = \int [(u')^2 + 2u^3].$$

They are conserved as formal coefficients; convergence or an analytic monodromy asymptotic theorem is not claimed.

Exercise 27.3 (Bäcklund transformation and a sine-Gordon kink). On $\mathbb{R}^{1,1}$ use $\xi = (x+t)/2$, $\eta = (x-t)/2$. For two real C^2 fields $\phi, \widetilde{\phi}$, put $\alpha_B = (\phi + \widetilde{\phi})/2$ and $\beta_B = (\phi - \widetilde{\phi})/2$. For a fixed $a_B > 0$, the proposed *Bäcklund system* is

$$\partial_\xi \alpha_B = a_B \sin \beta_B, \qquad \partial_\eta \beta_B = a_B^{-1} \sin \alpha_B.$$

(a) Prove that any such pair solves $\phi_{\xi\eta} = \sin\phi$ and $\widetilde{\phi}_{\xi\eta} = \sin\widetilde{\phi}$.

(b) Put $\widetilde{\phi} = 0$ and solve for the branch $0 < \phi < 2\pi$ at finite points. Convert it to physical coordinates and find its velocity.

(c) For density $L = \tfrac{1}{2}(\phi_t^2 - \phi_x^2) - (1 - \cos\phi)$, compute the kink charge $Q = [\phi(+\infty) - \phi(-\infty)]/(2\pi)$, energy and lower-index momentum integral $\int \phi_t \phi_x \mathrm{d}x$. This line solution is not a periodic field for the monodromy theorem.

Hint. Differentiate both first-order equations once across the other variable. For the vacuum partner integrate $\log\tan(\phi/4)$. In the energy integral use the rest-profile derivative $2\operatorname{sech} z$.

Answer. The branch is

$$\phi(t,x) = 4\arctan e^{\gamma(x - v_{\mathrm{tr}} t - x_0)},$$
$$v_{\mathrm{tr}} = \frac{1 - a_B^2}{1 + a_B^2}, \qquad \gamma = \frac{a_B + a_B^{-1}}{2} = \frac{1}{\sqrt{1 - v_{\mathrm{tr}}^2}}.$$

Its charge is one, energy 8γ, and lower-index momentum $-8\gamma v_{\mathrm{tr}}$.